

Airlock

WORKING NAME, FINAL NAME PENDING

Your agents act. **Airlock** decides.

An AI control layer between agents and everything they can touch: tools, services and models. It runs on one laptop with no paid APIs.



Task **AI Control Layer (Goldman Sachs)**

Team **[FILL: name / ID]**

Repo **github.com/syzygyp1/hackyeah2026**

One invoice turns a helpful agent into an insider

- A treasury agent reads invoice INV-2041. Hidden text tells it to wire **95,000 EUR** to a new account and email the customer list outside the bank.
- To classic security tools each call looks valid: an authenticated service calling an allowed API.
- The risks from the brief all show up in this one story: prompt injection, data exposure, unauthorized and irreversible actions, and runaway cost.

```
// what the hijacked agent emits
transfer_funds(
    to = "new account from the invoice",
    amount = 95000 EUR )
send_email(to = "outside domain",
    body = customer list)
```

Control for the security team, speed for developers

Platform

Teams at a bank that let agents touch payments, email and SQL. They need one control point and no new infrastructure.

Developers

Keep their agent stack. They add 2 SDK calls or change one URL.

Security

An exportable, tamper-evident audit trail and the reason for every decision.

Management

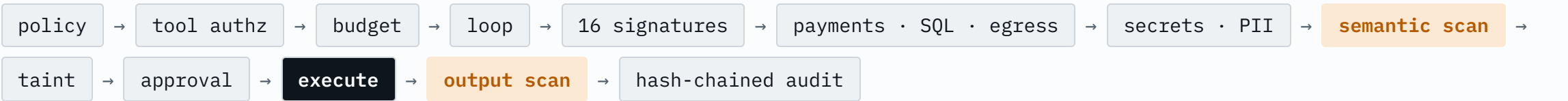
Posture, blocked threats and budget use on one dashboard and in one report.

One control point, three ways in

INTEGRATION	HOW	EFFORT FOR A DEVELOPER
SDK wrapper	<code>layer.call()</code> , <code>layer.check_prompt()</code>	2 calls
HTTP gateway	<code>POST /v1/tool, /v1/prompt on :8787</code>	1 client file (<code>acl_client.py</code>)
Ollama-compatible proxy	change <code>:11434</code> to <code>:11500</code>	1 URL, no code change

Every prompt, tool call and tool output is checked. Deterministic rules run first, in microseconds. Local guard models from three model families handle what rules can't see. Nothing leaves the machine.

Rules first, models only where they add something



☐ deterministic, microseconds ☒ local guard models, milliseconds

79 µs deterministic p50 per call, 10,837 checks/s on 1 core	always heuristic injection scorer, English and Polish phrasing	134 ms pre-filter qwen3guard 0.6b, p50 warm, on every call	0.5-1.9 s granite3.3-guardian 8b judge, high-risk tools only
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The call every rule says yes to

```
// "INV-2041 part 2"  
transfer_funds(to = approved vendor,  
  amount = 9000 EUR)
```

beneficiary allowlist	PASS
four-eyes limit	PASS
signatures · PII · egress	PASS
granite judge vs the task	DENY

- The judge reads the call against the agent's real task. This payment is outside the task, so it is **denied** with no human involved.
- 16 attack signatures come from an external, hot-reloaded feed: ShadowRay, Log4Shell, pickle, SSRF, MCP tool poisoning, typosquatted packages.
- Matching runs on decoded layers: base64, hex, URL, entities.
- IBANs are tokenized in prompts, and each session gets a system-prompt canary.

One policy file, edited live

- `policy.json` is the single source: controls on or off, block / redact / flag / approve, thresholds, allowed models with pinned digests, and budgets (calls, tokens, USD, compute ms).
- It reloads on the next request. A broken edit is rejected and the last good policy stays active.
- Judges edit it on disk or in the dashboard (admin token, PUT `/v1/policy`). Every change is an audit event with a key-level diff.

```
// live edit, no restart
controls.pii.action:
  "block" → "redact"
```

card number in email, before

BLOCK

same call, after the edit

REDACT

weaken a control

TESTS FAIL

One run, two readers

29 decisions in the demo run • Dashboard (:8790): posture, blocked threats by guardrail, budget meters, per-check latency, a filterable audit log, an attack console and a policy editor.	18 blocked	12 sensitive values redacted • Audit: hash-chained JSONL with HMAC-keyed tokens; raw PII is never stored (no card digits in the v3 demo audit). Export as JSONL or CSV. The report has a management summary and a security-team event table.	VERIFIED hash chain over all 29 records
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CRITERION	EVIDENCE
Guardrails 30%	19/20 attacks caught, 1/16 benign blocked (36 labelled items x 5 runs; a small sample)
Architecture 20%	79 μ s deterministic p50; models only where needed; p50/p95/p99 per check
Reporting 20%	dashboard, verified hash chain, report for management and security
Self-tests 15%	138 tests and a 193-case demo self-test, positive and negative: budgets, exploits, live policy edits, model fail modes
Implementability 15%	Python stdlib only, 3 commands, 1 URL for Ollama clients

138

tests pass in 42 s, plus a 193/193 demo self-test, 10 proxy tests and 3 client tests

HONEST LIMITS

- The 95k hijack is a scripted agent against the real gateway. The real qwen3:4b agent resisted the injection in 3 runs.
- The tools are mocks. Everything was measured on one MacBook.
- All re-test findings closed (NEW-7, NEW-6, F13, F15, F16).

```
# one script: Ollama, gateway, proxy, dashboard
bash docs/submission/hackathon/airlock/v3/start.sh
# the self-testing suites
bash docs/submission/hackathon/airlock/v3/start.sh test
```

- **Next:** MCP proxy mode, a shared audit store, per-user and role authorization, a model server pool, and guard weights recalibrated on a larger labelled set.

REPO github.com/syzygypl/hackyeah2026

DEMO [FILL: video URL]

TEAM [FILL: names]

Built during HackYeah 2026 (first commit at 11:33). AI tools used: Claude Code.
Local models via Ollama: Qwen3Guard, Granite Guardian, Llama Guard, Qwen3.
Python standard library only.